Milk Intake and Total Dairy Consumption: Associations with Early Menarche in NHANES 1999-2004

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Background Several components of dairy products have been linked to earlier menarche. Methods/Findings This study assessed whether positive associations exist between childhood milk consumption and age at menarche or the likelihood of early menarche (<12 yrs) in a U.S sample. Data derive from the National Health and Nutrition Examination Survey (NHANES) 1999–2004. Two samples were utilized: 2657 women age 20–49 yrs and 1008 girls age 9–12 yrs. In regression analysis, a weak negative relationship was found between frequency of milk consumption at 5–12 yrs and age at menarche (daily milk intake $\beta = -0.32$, $P<0.10$; "sometimes/variable milk intake" $\beta = -0.38$, $P<0.06$, each compared to intake rarely/never). Cox regression yielded no greater risk of early menarche among those who drank milk "sometimes/varied" or daily vs. never/rarely (HR: 1.20, $P<0.42$, HR: 1.25, $P<0.23$, respectively). Among the 9–12 yr olds, Cox regression indicated that neither total dairy kcal, calcium and protein, nor daily milk intake in the past 30 days contributed to early menarche. Girls in the middle tertile of milk intake had a marginally lower risk of early menarche than those in the highest tertile (HR: 0.6, $P<0.06$). Those in the lowest tertiles of dairy fat intake had a greater risk of early menarche than those in the highest (HR: 1.5, $P<0.05$, HR: 1.6, $P<0.07$, lowest and middle tertile, respectively), while those with the lowest calcium intake had a lower risk of early menarche (HR: 0.6, $P<0.05$) than those in the highest tertile. These relationships remained after adjusting for overweight or overweight and height percentile; both increased the risk of earlier menarche. Blacks were more likely than Whites to reach menarche early (HR: 1.7, $P<0.03$), but not after controlling for overweight. Conclusions There is some evidence that greater milk intake is associated with an increased risk of early menarche, or a lower age at menarche.